

IRBEYSKOYE, Russian Federation

WMO: 295870

Lat: 55.6636N

Lon: 95.4975E

Elev: 827

StdP: 14.26

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-45.4	-37.3	-47.1	0.4	-43.7	-41.1	0.5	-36.4	19.5	16.4	17.6	19.6	0.5	230	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.4	84.5	65.9	81.0	64.2	78.0	63.4	68.6	79.4	66.7	77.1	64.9	74.9	4.4	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.0	95.3	73.1	63.0	88.7	71.0	61.1	82.8	69.2	33.4	79.5	31.8	77.5	30.4	74.9	75.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.6	14.4	12.2	DB	-45.9	91.5	6.2	2.3	-50.3	93.2	-54.0	94.6	-57.5	95.9	-62.0	97.6	(4)
(5)			WB	-45.3	71.8	5.3	2.5	-49.1	73.6	-52.2	75.1	-55.1	76.5	-59.0	78.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	31.6	-7.0	1.7	20.7	35.9	47.5	60.5	64.2	60.3	46.3	32.1	11.5	3.7		
	DBStd	26.70	18.34	15.47	12.52	7.32	8.45	7.18	4.62	5.83	7.34	9.05	16.74	17.10		
	HDD50	7921	1767	1353	909	426	153	9	0	2	157	555	1154	1436		
	HDD65	12291	2232	1773	1374	873	545	168	71	170	561	1019	1604	1901		
	CDD50	1208	0	0	0	3	75	323	441	320	46	0	0	0		
	CDD65	104	0	0	0	0	2	32	47	23	0	0	0	0		
	CDH74	1651	0	0	0	3	76	681	531	336	24	0	0	0		
	CDH80	385	0	0	0	0	10	188	109	77	1	0	0	0		
	WSAvg	4.4	3.5	4.0	4.9	5.6	5.5	3.9	3.6	3.5	4.1	4.6	4.7	4.4		
	PrecAvg	18.3	0.6	0.5	0.5	1.1	1.7	2.4	2.9	3.4	2.1	1.3	1.1	0.9		
(15) Precipitation	PrecMax	23.0	1.0	1.1	1.3	1.9	3.6	4.2	7.3	6.3	3.3	2.0	2.0	1.8		
	PrecMin	11.5	0.2	0.2	0.2	0.1	0.5	0.8	0.4	1.7	0.3	0.4	0.5	0.3		
	PrecStd	3.0	0.2	0.2	0.3	0.5	0.8	0.9	1.6	1.3	0.9	0.5	0.4	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.0	37.8	51.6	69.7	81.7	89.1	87.7	86.6	77.9	60.2	47.6	36.9		
		MCWB	30.1	31.9	41.0	50.4	58.7	67.3	68.1	68.1	61.0	47.2	40.2	31.8		
	2%	DB	29.3	33.2	45.1	61.7	76.4	85.1	83.1	81.4	69.8	53.5	41.4	32.3		
		MCWB	26.3	28.7	37.3	47.1	56.4	64.9	66.0	66.6	57.6	45.0	36.3	28.4		
	5%	DB	25.2	30.2	41.4	56.3	71.0	81.4	79.9	77.5	64.4	48.6	37.4	29.4		
		MCWB	22.4	26.7	34.8	43.9	53.8	63.8	65.5	64.1	54.8	42.2	33.4	26.3		
	10%	DB	20.4	26.3	37.9	51.5	66.2	77.8	76.8	73.7	59.3	44.7	32.9	26.3		
		MCWB	18.2	23.5	32.6	41.7	51.9	62.4	64.2	62.2	52.1	39.8	30.3	23.9		
	0.4%	WB	30.4	32.3	41.9	51.6	61.5	70.2	71.3	70.8	61.4	49.4	41.1	32.3		
		MCDB	34.3	37.2	50.1	67.8	77.2	82.9	82.2	82.2	73.9	56.2	46.7	36.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	26.7	29.2	37.9	48.2	58.4	67.5	68.7	67.8	58.7	45.8	36.7	28.9		
		MCDB	29.4	32.6	44.3	59.9	72.2	80.0	78.5	78.3	68.3	52.4	41.3	31.9		
	5%	WB	22.6	26.6	35.4	44.9	55.8	65.7	66.9	65.2	56.1	43.0	33.3	26.5		
		MCDB	25.1	29.8	40.6	54.5	67.9	77.4	76.3	75.3	63.3	47.8	36.5	29.1		
	10%	WB	18.2	23.2	32.9	41.9	53.1	63.7	65.3	63.3	52.7	40.3	30.6	24.1		
		MCDB	20.4	26.4	37.3	50.9	64.2	74.9	74.4	72.2	58.5	44.4	32.6	26.4		
	5% DB	MDBR	20.7	24.6	23.5	20.4	22.7	24.9	20.4	20.7	20.0	17.1	17.5	17.9		
		MCDBR	21.2	24.5	26.3	32.2	33.9	34.0	28.0	29.8	31.8	25.0	17.5	16.2		
		MCWBR	18.5	20.4	18.6	18.7	16.8	15.3	12.8	15.0	18.6	16.3	12.1	13.9		
		MCWBR	20.0	22.9	23.7	29.1	28.6	28.8	23.8	27.2	28.9	23.8	17.4	15.5		
(40) Clear-Sky Solar Irradiance	taub		0.279	0.314	0.356	0.394	0.414	0.409	0.428	0.407	0.365	0.346	0.312	0.281		
		taud	2.028	2.004	1.981	2.134	2.183	2.296	2.262	2.288	2.356	2.257	2.090	1.973		
	Ebn at Noon		189	228	249	260	262	266	257	254	250	220	174	151		
		Edh at Noon	21	32	42	42	43	39	39	36	29	25	20	18		
(44) All-Sky Solar Radiation	RadAvg		260	581	1038	1381	1571	1882	1696	1374	892	493	252	162		
	RadStd		25	34	72	78	154	160	131	117	102	37	29	15		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (15 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page